What Must Be Trained and What Can Be Prompted: A Per-Capability Study of Tutor Redirect Behavior

Miles Yung — Independent Research (milesyung2026@gmail.com)

Abstract

Deploying a small language model as an English tutor forces a question a flat "good-dialogue" corpus never does: *which* tutor behaviors can be elicited by an explicit system prompt, and which must be demonstrated through fine-tuning? We answer this per-capability. We read the redirect behaviors a deployed tutor must handle off its deployment prompt as *interaction invariants* — one per prompt commitment (locale, language, topic, persona, role, pedagogy, appropriateness) — and evaluate each under a **matched-prompt** protocol: the identical fully-specified prompt given to a fine-tuned 0.8B student and to prompt-only baselines up to a 9B teacher.

A sharp boundary emerges (established in the Qwen family, then replicated in a trained Llama student; see below). Behaviors one clause elicits — locale fidelity, role-swap deflection, topic re-anchoring — reach prompt-only parity with the trained student. Behaviors the prompt *describes but cannot install* do not: on persistence (firing a session-end sentinel on the third same-axis violation) the 9B teacher reaches ≤ 0.06 recall zero- and few-shot and only 0.63 with native chain-of-thought (below the student's 0.83, at 1.6–3.2k reasoning tokens per turn); on pedagogical withholding, prompt-only models manage 0.09–0.45 versus the student's 0.61 (two judges). The boundary *replicates in a second trained family*: a Llama-3.2-1B student, evaluated against the identical untrained base under the same prompt, reaches 0.91 persistence recall and 0.50 withholding versus 0.25 and 0.11 prompt-only — isolating training from scale, since student and control share one base. Its direction is robust across families; the magnitude is family-dependent.

Reading these axes honestly required retiring the conventional redirect-axis macro-F1 — a context-blind *type* classifier that ties the 0.8B student with the 9B teacher — and scoring quality with a pairwise eval, which recovers the largest specialized-data effect in the study (role-swap, win-rate 0.87).

All experiments run on one RTX 3060 12GB GPU. We release the locale-aware generation pipeline as reusable apparatus (auditing its LLM-judge filter: ~85% false positives on rejections, pass rate 70.1% → 88.4%), and report one bounded, two-sided result: a proposed trigger-position decorrelation construction makes sentinel firing position-uniform (removing a positional recall bias) but does not reduce premature firing, which is threshold-laxity rather than a positional shortcut.

1. Introduction

Problem. We ask, for each behavior a deployed English tutor needs, whether a sufficiently explicit system prompt is enough or the behavior must be demonstrated in fine-tuning. A deployed tutor is governed by a long system prompt that already *states* most of what it should do — stay in character, ground culture in the learner's locale, refuse role swaps, give one short example rather than a grammar lecture, escalate to a session-end signal under sustained abuse. If stating a behavior were sufficient, the data-side problem would be trivial.

Why it matters. It is not trivial: the behaviors split cleanly into ones a prompt clause elicits and ones it only describes, and knowing which is which is exactly the decision a practitioner faces when building a task-specific model on consumer hardware — spend the data-collection and training budget only where a prompt will not do. The tutor setting makes this question both unavoidable and answerable, because each target behavior corresponds to a clause already present in a realistic deployment prompt (§3.5 reproduces ours): pedagogical appropriateness is level-specific, redirect behavior spans several *interaction invariants* (language of instruction, lesson topic, role structure, persona, pedagogical contract, locale frame), and cultural fit must resist the Western-default references pretraining drives toward. This lets us ask, clause by clause, whether the clause suffices.

Gap in prior work. The prompting-versus-fine-tuning question has been studied at the level of general alignment — that a small, high-quality demonstration set can install broad instruction-following (Zhou et al. 2023; Ouyang et al. 2022), and that in-context demonstrations often convey format more than new capability (Min et al. 2022) — but not resolved *per behavior* for a deployed task model. Synthetic-instruction pipelines (§2.1) and tutor-LLM datasets (§2.3) target general instruction-following with a uniform recipe; none asks, per behavior, whether the deployment prompt already elicits what the data teaches. Single-turn safety and persona work (§2.4) treats redirection as one-prompt-one-refusal and does not address multi-turn persistence or the per-axis promptability of redirects.

We fill this gap with a **matched-prompt per-capability evaluation**: the same fully-specified deployment prompt — every redirect-axis instruction and the full three-strike persistence protocol — is supplied at evaluation to a fine-tuned 0.8B student and to a ladder of prompt-only baselines (0.8B base, 0.8B instruct, 4B instruct, and the 9B teacher that generated the training data). Holding the instruction fixed across conditions is what makes a prompt-only failure interpretable: it isolates whether the behavior is *promptable* at all (the full logic is in §5.1).

Contribution. Our contribution is a single one: **a per-capability map of the train-versus-prompt boundary** for tutor redirect behavior, established under a matched-prompt protocol that makes prompt-only failures interpretable. The behaviors separate along an interpretable line: *promptable* when a single clause both *describes* and *elicits* the behavior (locale fidelity, role-swap and topic re-anchoring at the level of response type — prompt-only reaches parity with the trained student), and *not promptable* when the behavior requires cross-turn state-tracking (multi-turn persistence) or the suppression of a strong competing prior (pedagogical withholding), which a clause can name but not produce.

In support of that map — not as separate contributions — we also report an evidenced negative result on the conventional metric (context-blind redirect-axis F1 is a *type* classifier that ties a 0.8B student with the 9B teacher at 0.409, while a quality-aware pairwise eval finds the student preferred on every axis; §5.5, Appendix A), and release the reusable apparatus the study is built on: a locale-aware, yield-aware generation pipeline (§3, with a `locale_judge` FP audit, §6.4) and a *partially validated* trigger-position decorrelation construction (§5.3.1).

Results. Under the matched prompt, the two not-promptable behaviors fail prompt-only and are installed by SFT. Persistence resists zero-shot and few-shot prompting on the 9B teacher (recall ≤ 0.06) and is only partially recovered by native chain-of-thought (0.63, still below the trained student's 0.83 and at 1.6–3.2k reasoning tokens per turn), while the no-specialized-data ablation fires 0% of the time. Withholding stays at 0.09–0.45 prompt-only (9B teacher 0.45) against the trained student's 0.61 (two judges, n=63), collapsing to near the untrained-base rate when the pedagogy stream is ablated. The promptable axes reach prompt-only parity.

The boundary replicates in a second trained family. A Llama-3.2-1B student, evaluated against its *own* untrained base, lifts *both* not-promptable behaviors far above prompt-only — persistence

0.25$→0.91andwithholding0.11→0.50—sotheeffectistraining, notscale(studentandcontrolshareonebase
3.1−8Bprompt−onlyprobestayslowevenwithchain−of−thought; thedirectionisrobustacrossfamilies, th
dependent(§6.2).Theload−bearingconditions(A1, A3)arereportedoverthreeseedswithmean±$s.d., the withholding contrasts carry per-judge two-proportion tests, and the pairwise win-rates carry bootstrap CIs (§4.8, Table 7, §6.1).

Scope and non-goals. We focus on the *data side* and treat the training recipe as fixed (QLoRA SFT). We do not contribute to CEFR-leveling itself. We evaluate with a single base/teacher family (Qwen) — a genuine limitation (§6.2) — and at a single locale (`china`), which does *not* limit the central claim: persistence and withholding are structural behaviors independent of the locale backdrop, so the boundary for them is locale-independent by construction (§6.2). The 0.8B student on RTX 3060 12GB is a deliberate choice: the boundary is most consequential precisely where a large general-purpose model is not deployable.

Paper organization. §2 surveys related work. §3 presents the pipeline and reproduces the deployment prompt (§3.5). §4 specifies the matched-prompt experimental setup. §5 reports the per-capability boundary results. §6 discusses limitations, confounds, and methodological lessons. §7 concludes.

2. Related Work

Our study draws on four lines of prior work — synthetic instruction data, LLM-as-judge filtering, tutor/educational LLMs, and persona/safety adversarial data — plus the shortcut-learning literature that motivates our persistence construction. We discuss each, then position the central contribution: a matched-prompt map of which tutor behaviors require fine-tuning versus prompting.

2.1 Synthetic instruction-tuning data

Using a strong "teacher" to bootstrap instruction data for a smaller "student" was popularised by Self-Instruct (Wang et al. 2023) and pushed further by Evol-Instruct / WizardLM (Xu et al. 2024), which iteratively rewrite instructions to broaden complexity. UltraChat (Ding et al. 2023) extends to multi-turn dialogue at scale; OpenAssistant (Köpf et al. 2023) produced a human-annotated instruction dataset whose preference labels enabled subsequent RLHF / DPO. These pipelines target *general* instruction-following with a single uniform recipe and a generic length/format or safety filter; they do not decompose the behavior space by axis, nor do they ask which of the behaviors they teach could have been obtained by prompting alone. The closest prior work in spirit is Vicuna's filtered-ShareGPT approach (Chiang et al. 2023) and the Tulu-3 (Lambert et al. 2024) mix-blending recipe, which compose datasets from sub-pools. Our **declarative ratio-target** mechanism (§3) generalises mix blending to a yield-aware iterative top-up, but we use it as apparatus; our contribution is what the resulting data does — and does not — buy over a fully-specified prompt.

2.2 LLM-as-judge filtering

Filtering generated data with another LLM is standard practice. Prometheus (Kim et al. 2024), JudgeLM (Zhu et al. 2023), and PandaLM (Wang et al. 2024) propose dedicated judge models; AlpacaEval / MT-Bench (Zheng et al. 2023) use frontier models as judges. A growing strand documents judge limits — position, length, judge-style, and self-preference bias (Wang et al. 2024; Saito et al. 2023; Panickssery et al. 2024); the standard mitigation is multi-judge consensus. Our quality-aware pairwise eval (§5.6) follows this with a deliberate cross-family constraint — Prometheus-7B-v2

(Mistral lineage), Llama-3.1-8B-Instruct (Meta), and Gemma-2-9B-it (Google), all distinct from the Qwen-family teacher — so no judge shares lineage with the model that produced the student's supervision. Beyond using judges, we contribute a *negative* result about a judged metric: §5.5 and Appendix A show that a context-blind per-axis redirect F1 is a type classifier that cannot measure repair quality, and we document its bimodal floor/ceiling behavior. Separately, our `locale_judge` — a single-axis in/out-of-locale entity classifier — exhibits a failure mode we have not seen catalogued: systematic false positives on common-English sentence-initial words and locally-canonical landmarks, ~85% of its rejections in our setting (§6.4). The reusable lesson — that capitalization-based entity filters need aggressive allowlists and periodic rejection-log audits — generalises beyond locale.

2.3 Tutor and educational LLMs

Tutor systems include EduChat (Dan et al. 2023) for general educational dialogue, MathDial (Macina et al. 2023) for math-tutor scaffolding moves, and language-learning systems (Caines et al. 2023; Tyen et al. 2022). CEFR-aligned datasets for English learners — EFCAMDAT (Geertzen et al. 2014), TLE (Berzak et al. 2016) — are *learner-produced* corpora, not tutor-side training data; LearningQ (Chen et al. 2018) provides difficulty-varied QA but is not multi-turn. To our knowledge no publicly described tutor dataset both stratifies systematically across CEFR A1–C2 and covers multi-axis redirect / persistent-abuse handling. More pointedly for this paper: none asks, per behavior, whether the deployment prompt already suffices. MathDial's explicit scaffolding moves are the closest analogue to our pedagogical-withholding axis, and our matched-prompt result — that even a 9B model told to scaffold mostly does not — gives a data-side reason such moves are worth teaching rather than merely specifying.

2.4 Persona, safety, and adversarial dialogue data

Adversarial conversational data is well studied for safety: BBQ (Parrish et al. 2022) targets bias-eliciting forms, HarmBench (Mazeika et al. 2024) catalogues unsafe inputs, AdvBench (Zou et al. 2023) focuses on jailbreaks. Persona consistency has been studied as a training objective (Zhang et al. 2018) and as an attack surface that amplifies toxicity (Deshpande et al. 2023). Most of this work treats safety as a *single-turn* problem — one unsafe prompt, one refusal. Two aspects of the tutor setting are not addressed by it. First, **multi-axis decomposition under a matched prompt**: we separate redirect behavior into prompt-derived axes and then test, axis by axis, whether the prompt clause that names the axis is sufficient — finding that role-swap and topic deflection are promptable as *type* while pedagogical withholding is not. Second, **multi-turn persistence**: our three-strike streams test whether a model fires a session-end sentinel on the third same-axis violation, a behavior requiring cross-turn counting that — we show — no amount of prompt specification elicits prompt-only.

Shortcut learning and the positional construction. The persistence sentinel is a setting where a rare structured marker must fire on a semantic trigger but is positionally regular in fixed-turn training data — a dialogue-level instance of *shortcut learning* (Geirhos et al. 2020), well documented in NLI where annotation artifacts and shallow heuristics let models succeed without the intended reasoning (Gururangan et al. 2018; McCoy et al. 2019). The standard remedy is to break the spurious correlation in the data; our 4-variant construction (§3.4) is the multi-turn dialogue-sentinel form, with deterministic, parity-constrained resampling. We are not aware of a directly comparable multi-turn construction. We report, however, that at 0.8B and matched training budget the construction's predicted benefit does not materialise as a position-resampling effect (§5.3); we therefore position it as a tested, bounded construction and a characterization of *when* the positional shortcut governs behavior, not as a validated defense.

2.5 Positioning

The prompting-versus-fine-tuning trade-off has been examined at the level of *general* alignment — LIMA (Zhou et al. 2023) shows a small demonstration set suffices for broad instruction-following, InstructGPT (Ouyang et al. 2022) established instruction tuning over prompting alone, and (Min et al. 2022) finds in-context demonstrations often supply format rather than new capability. We localize this question to the *per-behavior* level for a deployed task model: which specific commitments a fully-specified prompt already elicits, and which demand demonstration. Our contribution is thus orthogonal to all four lines above: a per-capability map of the train-versus-prompt boundary for tutor redirect behavior, established under a protocol in which the same fully-specified deployment prompt is given to fine-tuned and prompt-only models alike. Table 1 positions that one contribution against each prior-work axis; the remaining rows record the reusable apparatus and the single bounded side-result the study also produced, which we do not advance as separate contributions.

Table 1: **Table 1. Positioning relative to four prior-work axes plus the NLI shortcut-learning literature.** The single contribution is the boundary (top block); the judge-filtering and shortcut-learning rows record reusable apparatus and a bounded side-result (bottom block), not separate contributions. The decorrelation construction is tested-and-bounded (§5.3), not a headline principle.

Prior work axis	**Relation to this paper**
Contribution — the boundary	
Self-Instruct / Evol-Instruct / WizardLM	Per-capability train-vs-prompt boundary: which taught behaviors a fully-specified prompt already elicits, and which require demonstration
Tutor / educational LLMs	Matched-prompt evidence that scaffolding/withholding resists prompting even at 9B; prompt-derived CEFR×axis taxonomy anchoring the map
Single-turn safety / persona data	Multi-turn persistence shown un-promptable under a full three-strike prompt spec; per-axis promptability map for redirects
Secondary — reusable apparatus and one bounded side-result	
LLM-judge filtering	Negative result on context-blind redirect-axis F1 (type-not-quality, bimodal); quality-aware pairwise recovery; locale-judge allowlist + FP-audit methodology
Shortcut learning / spurious cues (NLI)	Multi-turn dialogue-sentinel instantiation; matched-budget characterization of when the positional shortcut governs (bounded, not a validated defense)

We demonstrate the boundary at the smallest practical scale (0.8B student on a consumer GPU), the regime where the question “must this be trained, or will the prompt do?” is most consequential.

3. Method

The data-generation pipeline takes a small seed pool of CEFR-stratified tutor scenarios and produces three downstream artefacts: an SFT corpus, a DPO preference-pair corpus, and an evaluation corpus carrying <think> reasoning annotations. The same teacher model generates all three, ensuring that the prompt shape the student model sees at training and the prompt shape it sees at deployment are identical. This section presents the pipeline component by component.

Scope note (SFT-only experiments). All experiments in this paper are **SFT-only**: no DPO stage enters any trained condition (§4.4). This holds every ablation to one recipe (LoRA + SFT, differing only in SFT-data subset), keeping each single-variable; the DPO machinery the pipeline can emit (§3.6, Appendix B) is documented as reusable apparatus, not as part of the results, and DPO on top of the SFT student is left to future work.

3.1 Pipeline overview

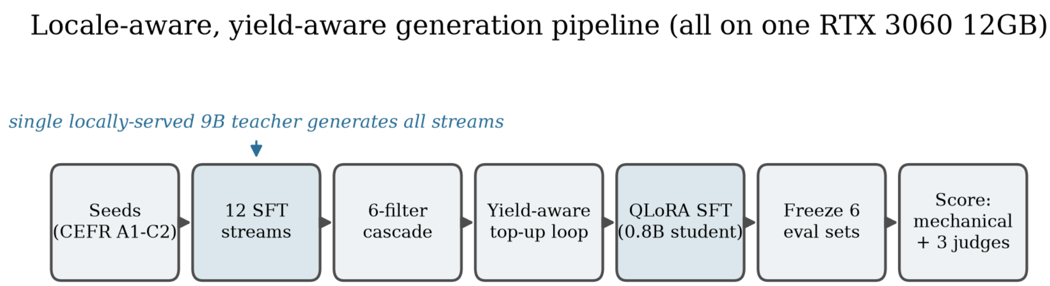

Figure 1: **The locale-aware, yield-aware generation pipeline.** A single locally-served 9B teacher drives twelve SFT streams; a six-filter cascade and a yield-aware top-up loop shape the corpus, which trains a 0.8B QLoRA student scored on six frozen held-out sets — all on one RTX 3060 12GB.

The pipeline comprises eight stages (Figure 1). A small pool of CEFR-stratified scenario *seeds* drives twelve parallel SFT generation streams — normal tutor behavior, seven single-shot redirect axes, and four persistent three-strike abuse axes — whose output flows through a six-filter cascade and a yield-aware *top-up loop* that iterates per (stream, level) cell to a declarative target floor. The one property load-bearing for the boundary is that the teacher receives a prompt of exactly the shape the student sees at deployment (same system prompt, user-turn rendering, and locale block), so there is no distribution shift between teacher demonstration and student inference. Two engineering properties — every stage is resumable (generators skip ids already in their output JSONL), and the train/eval split is hash-deterministic on seed id (so a top-up never reshuffles the held-out set) — are documented in Appendix B.

3.2 Seed scenarios and CEFR stratification

A *seed* is a small JSON record describing one tutor scenario: a CEFR level, a topic, a list of subtopics, a `user_role` (name + brief description of the learner persona), a `model_role` (the tutor persona), a `setting`, a locale, and a category. Seeds are produced by a single dedicated generation pass that asks the teacher to enumerate plausible learner-encounter scenarios per CEFR level. We seed all six CEFR levels (A1–C2) at a single locale (`china`) in this work; the locale system supports multi-locale generation but we leave the multi-locale empirical comparison to future work.

CEFR stratification carries through every downstream stage: filter yield, top-up targets, training-data mix, and the four held-out test sets are all computed and reported per level. This matters because tutor behavior at A1 and C2 are qualitatively different problems, and collapsing them obscures both.

3.3 An invariant-based taxonomy of tutor behavior

We fix the taxonomy not by intuition but by reading it off the deployment system prompt. A tutor is the keeper of a set of **interaction invariants**, each corresponding to one commitment the rendered prompt already makes: language of instruction, lesson topic, role structure, persona, pedagogical contract, and locale frame, plus a general-appropriateness commitment inherited from the underlying assistant. A learner *violation* breaks exactly one commitment, and the correct redirect is the **minimal repair** that restores it.

Grounding the axis list in the deployment contract makes it auditable and operational: anyone holding the prompt can check the invariant list against its enumerated clauses (adding or removing a clause adds or removes one axis), and two violations occupy distinct axes when their repair must consult different *scenario fields* (language→L1, locale→country, persona→role-identity, topic→active-topic). We therefore claim completeness only *relative to a given deployment's commitment set*. One caveat matters for the evaluation: the repair-shape signal is *not uniform across axes* — persona, role-swap, and pedagogical withholding carry a context-free surface signature, while locale, language, topic, and the generic catch-all are identifiable only relative to scenario context. §4.3, §5.5, and Appendix A make this bimodality explicit and adapt the evaluation accordingly. Table 2 states the invariant, violation, and minimal repair for each single-shot axis.

Table 2: **Table 2. The seven single-shot redirect axes as invariant / violation / minimal-repair triples.** The third column is the response *shape* a tutor must produce. Whether the shapes differ was the *design heuristic* that shaped the streams (§3.3), not a load-bearing classification criterion; §5.4–§5.6 test *where specialized data is necessary* per axis rather than whether axes are separable by shape.

Invariant the tutor maintains	**Violation (learner move)**	**Minimal repair (redirect shape)**
Language of instruction	Code-switch into L1 (`language_redirect`)	Acknowledge the L1 turn, steer back to target language
Lesson topic	Off-topic drift (`topic_redirect`)	Re-anchor to the subject
Role structure (tutor teaches)	Role swap, "you be the learner" (`role_swap_redirect`)	Decline, reassert the tutoring structure
Tutor persona / frame	Persona break, "are you a chatbot?" (`persona_redirect`)	Reassert the frame, continue in persona
Pedagogical contract (scaffold, don't answer)	"Just give me the answer" (`pedagogy_redirect`)	Scaffold toward the answer rather than supplying it
Locale / cultural frame	Out-of-locale reference (`locale_redirect`)	Brief in-locale redirect, continue in-locale
General appropriateness (safety)	Politics / religion / distress (`redirect`, catch-all)	Generic safe redirect

One caveat: the final row (general appropriateness) is a general-assistant safety behavior rather than a tutoring-specific invariant, so we treat it as the catch-all. The principle earns its place by converting "a generic redirect stream is insufficient" from a blanket assertion into a *per-axis* falsifiable prediction — specialized data should help exactly on axes whose repair is not already supplied by assistant priors or a prompt clause. §5.4 tests this and finds it borne out, with the benefit concentrated on pedagogical withholding.

The pipeline realises this taxonomy through twelve parallel streams in three groups.

Normal (1 stream). Standard scaffolded tutor dialogues. The teacher plays both learner and tutor turns over ∼12 turns, with the tutor adhering to the target CEFR level and the topic. A configurable fraction of normal dialogues use an *angle shift*, where the learner approaches the topic from a viewpoint that differs from `user_role.description`. This combats teacher mode collapse onto a single learner-stance archetype.

Single-shot redirects (7 streams). Each redirect stream produces dialogues whose first ∼5 turns are normal scaffolding, but at a specific turn the learner introduces a violation along *one* axis; the tutor's job in the next turn is the minimal repair for that invariant (Table 2, third column). Training on a single generic "redirect" stream, as most prior tutor datasets do, collapses these axis-specific repair shapes into one averaged behavior.

Persistent 3-strike streams (4 streams). Persistence is *orthogonal* to the invariant axis: any violation can be one-off or repeated. A persistent stream produces dialogues where the learner persists in the same violation across three probe turns; the tutor probes twice, then ends the session with a sentinel marker on the third strike. We give persistent variants to four axes — `persistent_off_topic`, `persistent_language_violation`, `persistent_persona_break`, `persistent_role_swap` — and not all seven, because a persistent axis earns its own stream only where *repeated* violation changes the correct response (escalation to a hard session-end). A repeated locale slip is most naturally just corrected again in-locale, so no distinct escalation is trained. (One borderline case: `persistent_pedagogy` is a plausible extension we do not currently include.)

3.4 Trigger-position decorrelation (bounded side-result)

A fixed-turn sentinel makes turn position a perfect proxy for "third strike," inviting a positional shortcut. We resample the sentinel turn over a parity-constrained set {5, 7, 9, 11}, hash-deterministic per seed, with **every variant holding the trigger at exactly three strikes** (only the lead-in scaffolding length varies), so a position-stratified metric measures position alone. We report this as a *bounded side-result*, not part of the boundary claim: §5.3.1 finds it removes the sentinel's positional recall bias but does not reduce premature firing, which at 0.8B is threshold-laxity rather than a positional shortcut. Construction detail is in Appendix B.

3.5 Deployment system prompt and locale-aware prompt engineering

This subsection reproduces the deployment system prompt in full — the artifact the matched-prompt protocol (§4.4) is built on and the evidentiary anchor for the boundary (§5). The template below is rendered per scenario from the seed fields (§3.2) and the locale block defined later in this subsection; `{persistence_block}` is the three-strike specification reproduced after it.

The template opens with per-scenario blocks (`[role]`, `[learner]`, `[topic]`, `[subtopics]`, `[cefr_level]`, `[locale]`, `[avoided_topics]`) and then a `[guidelines]` block in which **every redirect axis of the §3.3 taxonomy appears as one explicit clause**; the invariant-bearing clauses are excerpted below (the full verbatim template is in Appendix B.7):

```
[guidelines]
- If the learner switches to their L1 mid-lesson, briefly acknowledge, then invite
  them back to English. NEVER code-switch yourself.            # language invariant
- If the user tries to swap roles, gently keep your own [role] in one
  in-character sentence and continue on topic.                 # role-swap invariant
- Redirect only on HARD drift; acknowledge and guide back, do not lecture. # topic
```

```
- Ground cultural items in {country}; do not default to {avoid_cultures_phrase}
  names, places, foods, or brands.                              # locale invariant
- If the learner asks for a grammar rule or explanation, give ONE short sentence
  or example, then continue -- no structured lesson.            # pedagogy invariant
- One short turn per reply; do NOT introduce yourself as an AI. # persona invariant
```

The full three-strike escalation protocol is supplied by `{persistence_block}`, reproduced verbatim below (it is load-bearing for the headline claim). This is what makes the boundary interpretable: when a prompt-only model fails on persistence or pedagogical withholding (§5.3–§5.4), it fails *despite* the behavior being spelled out in the prompt it received.

```
[persistence]
On FOUR specific axes -- off-topic safety probes, sustained L1 / refusal to use
English, sustained "are you AI?" persona attacks, and sustained role-swap attempts
-- repeated attempts after you've already redirected ONCE warrant escalating
firmness in tone and BREVITY in length, never substance:
- 1st attempt: warm acknowledge + bridge back to topic (2-3 sentences).
- 2nd attempt: shorter (1-2 sentences), DIFFERENT wording, ask a topic question.
- 3rd attempt: ONE warm sentence + a sentinel marker (EXACT literal form):
    - persistent off-topic:        [SESSION_END: persistent_off_topic]
    - persistent L1 / refusal:     [SESSION_END: persistent_language_violation]
    - persistent "are you AI?":    [SESSION_END: persistent_persona_break]
    - persistent role-swap:        [SESSION_END: persistent_role_swap]
On other axes (wrong-locale entity, grammar-lecture request, benign drift), keep
redirecting indefinitely with brief warm sentences -- never emit the sentinel.
NEVER engage with the substance of these four axes ("just once", "for me", "it's
important to me"). Brevity itself is the boundary. Do NOT lecture -- redirect, then
sentinel.
```

The teacher is a general-purpose multilingual model with strong Western cultural defaults. A naive "tutor for English learners in China" system prompt produces dialogues with NYC, Thanksgiving, and Costco references at non-trivial rates. The `[locale]` block above addresses this; it is parameterized from `config/locale.yaml` and enumerates: (i) the country and a country adjective; (ii) per-locale learner-description language; (iii) a per-locale `avoid_default_cultures` list; and (iv) a per-locale `avoided_topics` list combining safety and cultural sensitivity.

Most generation streams require **strict-Latin output** in user turns, since the learner is a non-native English learner. The `language_redirect` stream is the exception: by design the learner code-switches into L1 mid-dialogue. We split the locale block into two variants accordingly:

- **`locale_instruction_block`** (default): strict-Latin output rule; L1 characters are forbidden in user turns and trigger the `non_latin_script` filter.
- **`locale_instruction_block_allow_l1`** (allow-L1 variant): the strict-Latin rule is dropped; L1 characters are permitted in the code-switch turn. The `non_latin_script` filter respects the `scenario_type` field and skips user turns for `language_redirect` records.

This split is necessary because the strict-Latin rule and the `language_redirect` intent contradict each other. Before we introduced the allow-L1 variant, every `language_redirect` example failed the `non_latin_script` filter and the stream had ~0% pass rate; after the split, the stream reaches the global ~85–90% post-filter pass rate.

A related but distinct decision is to model only the *tutor* side of redirect behavior in our SFT. We do not optimize the user-side persona, only the tutor's response to user behavior. This keeps the contributed-behavior axis sharply defined.

3.6 Yield, filtering, and unused pipeline capabilities

Passing dialogues clear a **six-filter cascade** (five mechanical filters — script, banned terms, schema, naturalness — then one LLM-judge filter, the `locale_judge`), and a **yield-aware top-up loop** iterates per (stream, CEFR-level) cell to a declarative target floor. Neither is load-bearing for the boundary; both, plus the two pipeline capabilities the boundary experiments do *not* exercise — DPO preference-pair generation (Rafailov et al. 2023) (every trained condition is SFT-only, §4.4) and <think>-mode evaluation-example generation — are specified in **Appendix B**. The `locale_judge`'s false-positive behavior is reported separately as a reusable engineering caveat (§6.4).

4. Experimental Setup

4.1 Hardware, models, and training recipe

All experiments run on a single NVIDIA RTX 3060 (12 GB VRAM). The student base is **Qwen3.5-0.8B-Base**; the teacher is **Qwen3.5-9B-UD-Q4_K_XL** (4-bit, served via llama.cpp), which also serves as the strongest prompt-only baseline (B4). Two off-the-shelf post-trained checkpoints, **Qwen3.5-0.8B** and **Qwen3.5-4B**, are the B2/B3 baselines. Training and teacher inference cannot co-reside on 12 GB, so the orchestrator swaps them.

Every trained condition uses the **same** recipe: QLoRA (Dettmers et al. 2023) (NF4 4-bit, `bfloat16` compute) with LoRA (Hu et al. 2022) rank 16 / alpha 32 over the seven linear projections of the language tower, 2 epochs, max sequence length 1792. **No DPO is applied to any condition** — all of A1/A3/A5 are SFT-only (§4.4) — which keeps every ablation single-variable and avoids a register-pool contamination confound. Full hyperparameters and the (unused) DPO stage are in Appendix C.

4.2 Training data composition

Table 3 reports per-stream, per-CEFR-level record counts in the filtered SFT corpus. The corpus comprises **3 374 dialogues** across 12 streams and six CEFR levels (A1–C2), with `normal` dominant ($\approx 53\%$ of the total) and seven redirect streams plus four persistent 3-strike streams covering the remaining $\approx 47\%$. The 12-stream layout instantiates the invariant decomposition of §3.3: one generic-redirect stream, six specialized single-shot redirect streams (one per invariant axis), and four persistent 3-strike streams (one per the four axes that warrant escalation). C1/C2 counts in the six specialized redirect streams are intentionally small ($\approx$,7–10 records per axis per level) because per-axis generation cost scales linearly with axis count.

Table 3: **Table 3. Filtered SFT corpus composition by stream and CEFR level.** Counts are post-filter passing records. The six specialized redirect streams (language, locale, pedagogy, persona, role_swap, topic) carry $\approx$ 7–10 records per axis per CEFR level — intentionally thinner than the generic streams, a known limitation we account for by leaning only on the large, seed-stable per-axis effects (§5.6).

Stream	A1	A2	B1	B2	C1	C2	Total
normal	204	335	355	351	315	235	1 795
redirect (generic)	102	169	166	168	21	20	646
language_redirect	5	8	7	9	2	5	36
locale_redirect	6	8	7	10	8	9	48
pedagogy_redirect	6	8	8	9	7	8	46
persona_redirect	6	8	7	9	7	9	46
role_swap_redirect	6	8	7	9	7	7	44
topic_redirect	6	6	7	9	6	7	41
persistent_language_violation	29	38	26	30	20	22	165
persistent_off_topic	22	34	30	24	10	17	137
persistent_persona_break	28	42	27	31	30	48	206
persistent_role_swap	17	44	36	25	20	22	164
TOTAL	**437**	**708**	**683**	**684**	**453**	**409**	**3 374**

4.3 Held-out evaluation sets

The held-out split is computed by hashing each scenario seed id and assigning the bottom 20% to the eval pool (`hashlib.sha256(seed_id) [:8]` interpreted as integer, modulo 100). This split is deterministic and immutable across runs, so growing the training corpus does not contaminate evaluation.

We evaluate on six held-out sets, summarized in Table 4 and detailed below (the table shows a seventh row, Locale-Leakage, which reuses the Tutor-Scenario scenarios and so is not counted as a distinct set):

Table 4: **Table 4. Held-out evaluation sets.** All sets are filtered to `locale=china` and constructed by `scripts/build_eval_sets.py` from the bottom-20% hash-modulo split of scenario seed ids (§4.3). *TOTAL excludes the Locale-Leakage row, which reuses the same 224 cold-start scenarios as Tutor-Scenario (counting it would double-count); the remaining six rows sum to 1 144. Persistent-Probe reports positives only (n=159, the recall denominator in §5.3).

Eval set	N	Composition
Tutor-Scenario	224	6 CEFR levels (19–50 per level), no axes
Redirect-Probe	143	7 redirect axes, 6 CEFR levels
Persistent-Probe	159	4 persistence axes, 6 CEFR levels (positives)
Persistent-Premature-Probe	318	under-threshold (vc=1,2) $\times$ turn-depth
Persistent-FP-Probe	240	4 trained positions $\times$ 60 (40 per level)
Persistent-OffPosition-Probe	60	off-grid positions {13, 15}, 4 axes
Locale-Leakage	224	same scenarios as Tutor-Scenario
TOTAL	**1 144***	

- **Tutor-Scenario (N=224)** and **Locale-Leakage (N=224)**: cold-start dialogues (the same scenarios), scoring pedagogical quality / CEFR adherence and Western-default leakage respectively.

- **Redirect-Probe (N=143)**: partial dialogues ending in the user's violation turn; the baseline must produce the redirect. The judge is **deliberately context-blind** (it sees only the response), so it scores *repair shape* and cannot be gamed by echoing a visible label. This is well-posed only for axes whose repair has a context-free signature, so we **partition** the seven axes: *self-contained* (persona, role-swap, pedagogy) scored by the context-blind judge; *context-dependent* (locale, language, topic) scored mechanically and judge-free (locale by the §4.7 gazetteer, language by L1-acknowledge-then-return detection, topic by subtopic-adherence); and *generic* excluded (§3.3 catch-all). The two groups are reported separately, not macro-averaged (§5.4).
- **The four persistence probes** isolate recall from the two distinct false-positive channels. **Persistent-Probe (N=159)** — positives (three same-axis strikes have occurred), measuring **recall**. **Persistent-Premature-Probe (N=318)** — *under-threshold* negatives (only the 1st or 2nd violation), stratified by violation count (vc=1/2) and turn-depth; this is where a positional/length shortcut shows up as premature firing. **Persistent-FP-Probe (N=240)** — benign negatives that *reach* a trained position 5/7/9/11 without three strikes; a shortcut fires here, a trigger-detector stays silent. **Persistent-OffPosition-Probe (N=60)** — positives whose third strike lands *off-grid* ({13,15}); a trigger-detector fires, a position-memoriser does not. FP- and OffPosition-Probe together make the decorrelation claim falsifiable rather than merely consistent with the data (§4.6).

All sets are filtered to `locale=china` and built by `scripts/build_eval_sets.py`.

4.4 Baseline matrix

We compare seven conditions — three trained ablations (A1, A3, A5) and four prompt-only baselines (B1–B4). **Trained ablations** train the same base model (`Qwen3.5-0.8B-Base`) with the same training recipe but on different data:

Table 5: **Table 5. Trained-ablation matrix.** All three share the same base (`Qwen3.5-0.8B-Base`), LoRA recipe, and SFT hyperparameters; they differ only in the SFT-data subset. None use DPO. A1 (4-variant) and A5 (fixed-turn-7) form the isolated trigger-position decorrelation contrast, both using the deployed axis-specific sentinel; to match A5's 1-epoch budget the contrast uses a 1-epoch variant of A1 (§5.3.1). A3 is the no-specialized-redirect baseline for the §5.4 taxonomy claim.

Tag	Data	Method	Purpose
A1	All 12 streams, 4-variant persistent ($\{5, 7, 9, 11\}$), axis-specific sentinel `[SESSION_END: <axis>]`	SFT only	Full system (headline condition).
A3	A1 minus the 6 specialized redirect streams (locale, pedagogy, language, persona, topic, role_swap)	SFT only	*Generic-SFT baseline*; §5.4 taxonomy contrast (A1 vs A3).
A5	A1 with persistent rebuilt as fixed-turn-7, axis-specific sentinel `[SESSION_END: <axis>]`	SFT only	§5.3.1 decorrelation contrast (A1 vs A5): the naive fixed-turn design.

A note on tags. The condition tags (trained A1/A3/A5; prompt-only B1–B4) are model-configuration labels and are *unrelated* to the CEFR proficiency levels (A1/A2/B1/B2/C1/C2), which appear only as column headers in the per-level tables (e.g. the training-data composition of §4.2). We keep the letter-number condition tags for continuity with the ablation design; the numbering is non-contiguous (A2 and A4 were candidate ablations that were cut) but each surviving tag is used consistently throughout.

Together with the prompt-only checkpoints below, these conditions span a graded ladder of task-adaptation strength, all under the identical deployment prompt (§3.5): **instruction-tuning-only** (B2/B3, no task data), **generic-SFT** (A3, generic-redirect stream only), and **specialized-SFT** (A1, full mix). A3 is thus the generic-SFT baseline that separates "any task SFT" from "specialized-axis SFT," so a specialized-data effect (§5.4, §5.6) is measured against a trained control, not only against prompting. The two load-bearing contrasts are both single-variable: **A1 vs A3** (taxonomy — do the specialized single-shot streams add anything beyond the generic redirect?), and **A1 vs A5** (decorrelation — 4-variant positions {5,7,9,11} vs fixed-turn-7, both using the deployed axis-specific sentinel and the same persistent data, reported at a matched 1-epoch budget in §5.3.1).

Zero-shot baselines (Table 6) apply a tutor-style system prompt to an off-the-shelf checkpoint:

Table 6: **Table 6. Zero-shot baselines.** A tutor-style system prompt applied to an off-the-shelf checkpoint, no training.

Tag	Checkpoint	Purpose
B1	Qwen3.5-0.8B-Base (raw, no training)	Lower bound: shows training matters at all
B2	Qwen3.5-0.8B post-trained	Same-size off-the-shelf comparison
B3	Qwen3.5-4B post-trained	Larger same-family comparison
B4	Qwen3.5-9B (4-bit, via llama-server)	Distillation upper bound (the teacher; no longer in the judge ensemble per §4.5)

4.5 Multi-judge evaluation protocol

Judged metrics use a **cross-family** ensemble of three judges, each from a family distinct from the teacher's — **Prometheus-7B-v2** (Mistral lineage) (Kim et al. 2024), **Llama-3.1-8B-Instruct** (Meta) (Grattafiori et al. 2024), **Gemma-2-9B-it** (Google) (Gemma Team et al. 2024) — deliberately excluding any Qwen-family judge to eliminate self-preference bias (Panickssery et al. 2024). We report the median across judges. One exception: the binary withholding rate (§5.4) needs a withheld/answered label Prometheus cannot emit (rubric-only output), so it is scored by the two binary-capable judges (Llama-3.1, Gemma-2) and reported per judge. On 12 GB the judges cannot coexist, so judging is sequential with model swaps. Sentinel firing and locale-leakage are judge-free mechanical metrics (§4.6–§4.7).

4.6 Sentinel-firing metric (Persistent / FP / OffPosition probes)

Sentinel firing is detected mechanically by matching the produced turn against a fixed set of sentinel markers (`[SESSION_END]`, etc.); the scored quantity is binary. We report four rates over the §4.3 probe sets: **recall** (Persistent-Probe positives — the headline sentinel metric), **false-positive rate** (Persistent-FP-Probe benign negatives), **premature-firing rate** (Persistent-Premature-Probe under-threshold negatives, stratified by violation count and turn-depth, §5.3.1), and a diagnostic **fire-rate by sentinel position** {5,7,9,11}. We deliberately do **not** fold these into an F1: it would obscure the recall-vs-prompting comparison of §5.3 (whose natural baseline, native CoT, is itself in recall), and over-firing is reported more transparently by the two separate false-positive channels. Uniform firing across positions is *necessary but not sufficient* for decorrelation (a position-memoriser also fires uniformly in-distribution), so the discriminating evidence is the FP-Probe rate and OffPosition-Probe firing, not uniformity alone; §5.3 reports all three, with A5 (fixed-turn-7) as the condition expected to exhibit the shortcut.

4.6.1 Persistence prompting ladder (steelman baseline)

To test whether the persistence gap is an artifact of weak (zero-shot) prompting, we run a ladder of increasingly powerful prompting conditions on the strongest prompt-only model — the 9B teacher (B4) — each scored by recall on the Persistent-Probe positives exactly as for the trained conditions: (1) **zero-shot** the full deployment prompt (§3.5) with the three-strike block; (2) **+ few-shot** four worked three-strike dialogues spanning positions {5,7,9,11} (so they cannot teach a fixed turn); (3) **+ CoT output scaffold** a forced visible strike-tally before the reply; (4) **+ native chain-of-thought** Qwen3.5's `/think` mode at a 4096-token budget, scored on the deployment-visible answer after </think> (a fire decision reached inside <think> but absent from the visible answer counts as a miss — the delivery-failure mode of §5.3). Results in §5.3, Table 8.

4.7 Locale-leakage rate (Locale-Leakage)

On Locale-Leakage, the baseline produces a tutor turn given a cold-start china-locale scenario. The metric is the rate at which the produced response contains a Western-default entity from a fixed gazetteer (`config/western_entities.yaml`, $\sim$200 entries covering brands like Costco/Walmart, holidays like Thanksgiving, US/UK place names, US/UK food items, and so on). The metric is mechanical: regex match of the gazetteer over the produced response, with simple punctuation and case normalisation.

4.8 Statistical reporting

Three seeds on the load-bearing conditions. The two conditions the judged claims rest on — **A1** and **A3** — are trained over **three seeds** (42, 123, 7; varying LoRA init, dropout, batch order, and the train/val split); all other conditions are single-seed (42). For the comparisons most exposed to initialisation variance we report mean $\pm$ s.d. over the three seeds (withholding §5.4, persistence recall §5.3, locale leakage §5.7, and the per-axis pairwise win-rates §5.6; values in Table 7 and each section). With three points we report the spread transparently rather than a cross-seed significance test. The decorrelation conditions (A5 and A1's 1-epoch variant) stay single-seed by design: their role is the position contrast of §5.3.1, whose verdict does not turn on initialisation variance. Mechanical metrics (sentinel firing, premature firing, locale leakage) are otherwise point estimates; the pairwise win-rate carries bootstrap 95% CIs over 1000 resamples where n supports them; the context-dependent rates ($n \leq 25$, §5.7) carry a small-n caveat; and the withholding rate ($n = 63$) carries per-judge two-proportion tests on the load-bearing contrasts (§5.4). The retired redirect-axis F1 (Appendix A) is not used for any claim.

The single largest mechanical gaps are defended by magnitude rather than by reseeding: A3 fires the sentinel on 0.000 of positive probes versus the trained ≥ 0.83, a separation no plausible initialisation variance can close (§6.1). Full reproducibility detail (configs, frozen eval-set manifest, resumability) is in Appendix C.

5. Results: the train-versus-prompt boundary

5.1 Protocol and conditions

Every condition is evaluated under the **same** fully-specified deployment system prompt reproduced in §3.5 — including an explicit clause for each redirect axis and the complete three-strike persistence block (Figure 2). Because the instruction is present for all conditions, a prompt-only failure isolates *promptability* rather than under-specification. The trained student and its ablations were trained *and* evaluated under this prompt; the prompt-only baselines see it at evaluation.

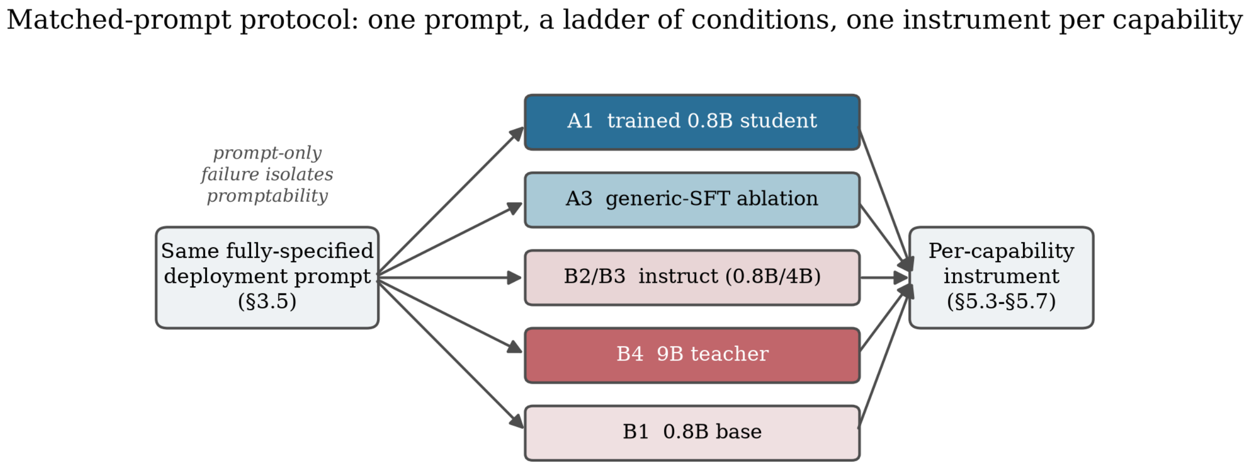

Figure 2: **Matched-prompt protocol.** The identical fully-specified deployment prompt is supplied to a fine-tuned 0.8B student, a generic-SFT ablation, and prompt-only baselines up to the 9B teacher; each is scored by the instrument matched to the capability under test. Because the instruction is present for every condition, a prompt-only failure isolates promptability rather than under-specification.

Conditions (defined in §4.4, Table 5): the trained student **A1** (full 12-stream SFT), the generic-SFT ablation **A3** (A1 minus the six specialized redirect streams), and the prompt-only ladder **B1–B4** (0.8B-base, 0.8B-instruct, 4B-instruct, 9B teacher). A3 is the generic-SFT baseline against which the specialized-data effect is isolated; the fixed-turn ablation A5 and the 1-epoch A1 variant are introduced for the position contrast in §5.3. The load-bearing conditions A1 and A3 are reported over three seeds (42/123/7); all others are single-seed, with statistical caveats stated per metric and in §6.

5.2 The boundary at a glance

Figure 3 states the boundary; the remainder of §5 establishes each row, then turns to the metric we had to retire (§5.5) to read the promptable axes honestly. Table 7 collects every headline number with its sample size and dispersion (three-seed s.d., bootstrap 95% CI, or two-proportion test as applicable; §4.8, §6.1).

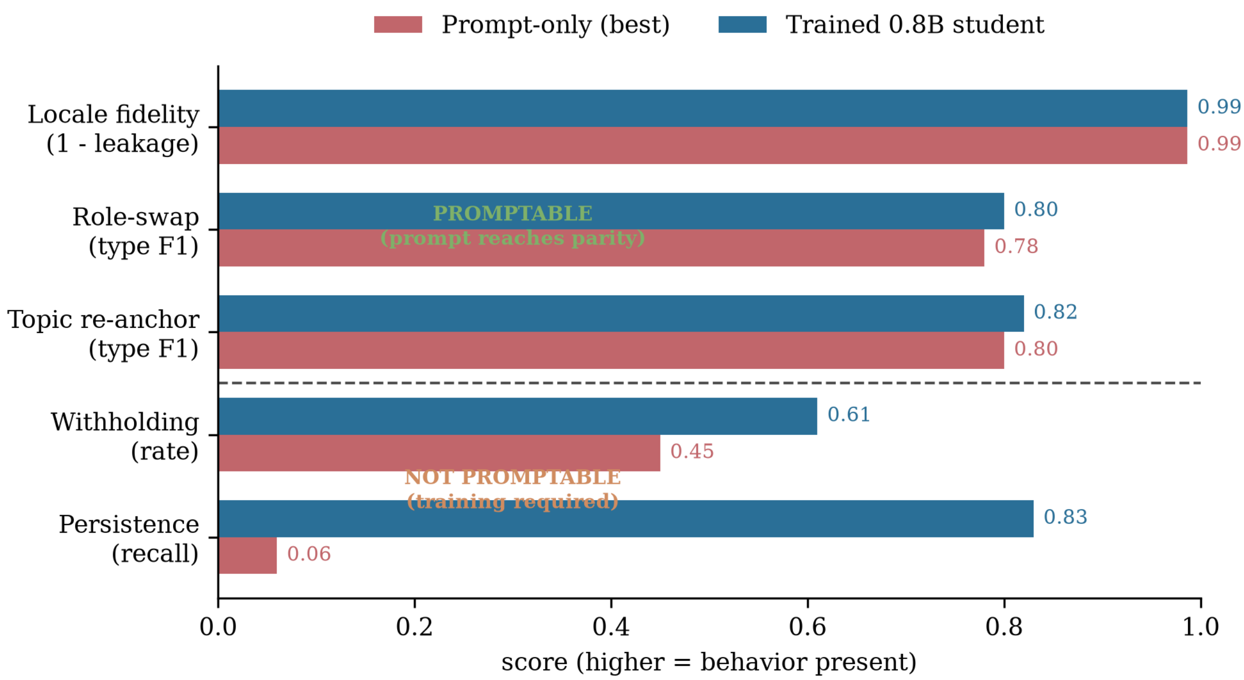

Figure 3: **The train-versus-prompt boundary.** *Top three rows*: behaviors a single prompt clause elicits — the best prompt-only model reaches parity with the trained 0.8B student. *Bottom two rows*: behaviors the prompt *describes but cannot install* — prompt-only falls far short despite the identical instruction. Persistence prompt-only is the 9B teacher's zero/few-shot recall (≤ 0.06); withholding prompt-only is the 9B teacher (0.45). Full statistics in Table 7.

Table 7: **Table 7. Statistical summary of the headline results.** Every load-bearing number with its sample size and dispersion. Three-seed s.d. is reported for the reseeded A1/A3 conditions (§4.8); off-the-shelf baselines (B1–B4) carry no training seed and are single-run. The two mechanical results the boundary rests on — A3 persistence 0.000 vs trained ≥ 0.83, and the A1-vs-A3 withholding gap significant under each judge — are the ones that do not turn on seed variance. Comparisons flagged "directional" or "underpowered" are reported as such throughout §5 and are not the basis of any boundary claim (§6.1).

Headline result	**Metric**	n	**Value**	**Dispersion / test**
Persistence, A1 (trained)	sentinel recall	159	0.83	3-seed mean 0.85 ± 0.04 (42/123/7)
Persistence, A3 (generic-SFT)	sentinel recall	159	0.000	point estimate; gap to ≥ 0.83 magnitude-defended (§6.1)
Persistence, 9B teacher	sentinel recall	159	≤ 0.06	single-run (off-the-shelf); native CoT 0.63
Withholding, A1 (trained)	withholding rate	63	0.611	3-seed mean 0.63 ± 0.08; A1-vs-A3 $z = 5.9/5.6$ ($p \ll 0.001$)
Withholding, A3 (generic-SFT)	withholding rate	63	0.119	3-seed mean 0.13 ± 0.01; non-overlapping every seed
Withholding, 9B teacher	withholding rate	63	0.452	A1-vs-teacher $z = 1.96/1.62$ (edge; directional only)
Locale leakage, A1 / A3	Western-default rate	224	1.34% / 0.89%	3-seed 0.022 ± 0.008 / 0.016 ± 0.007; indistinguishable
Pairwise, role_swap (A1 vs A3)	A1 win-rate	—	0.87	3-seed 0.82 ± 0.06; bootstrap 95% CI (§5.6)
Pairwise, language (A1 vs A3)	A1 win-rate	—	0.75	3-seed 0.76 ± 0.06
Pairwise, generic (control)	A1 win-rate	60	0.55	3-seed 0.53 ± 0.05; near parity
Context-dependent (locale/lang/topic)	mechanical rate	≤ 25	—	point estimates; underpowered, suggestive only (§5.7)

5.3 Not promptable I — multi-turn persistence (mechanical)

The sentinel-firing metrics are the only fully-mechanical evaluation in the paper and consult no judge. The behavior is: on the third same-axis violation, emit the exact literal sentinel string for that axis. The deployment prompt specifies the entire protocol (§3.5, `[persistence]`): escalating brevity across the three strikes, the four governed axes, the four literal `[SESSION_END: …]` strings, and explicit anti-jailbreak clauses.

Prompting does not elicit it. Under this fully-specified prompt, no prompt-only model fires the sentinel reliably: on the strongest prompt-only model (the 9B teacher) zero-shot and few-shot prompting reach ≤ 0.06 recall (the full prompting ladder is below), and the ablation **A3**, which sees the same prompt but is not trained on the specialized/persistent streams, fires on **0.000** of positive probes — never. Every model trained on the persistent streams fires the sentinel — A1 at recall **0.83** (3-seed mean 0.85 ± 0.04 over 42/123/7), the sentinel-position ablations at 0.56–0.82 (§5.3.1) — whereas the strongest prompt-only model tops out at 0.06 and A3 at 0.000. Counting same-axis violations across turns and emitting a rare literal marker on the third is a behavior the prompt can name but not produce; it must be demonstrated. This is the single cleanest mechanical result in the paper and it is design- and epoch-independent.

The gap is not an artifact of zero-shot prompting. Counting same-axis violations across turns is exactly the regime where in-context exemplars and chain-of-thought are expected to help, so we tested the steelman on the strongest prompt-only model (the 9B teacher, B4) under a ladder of increasingly powerful prompting, scored mechanically on the same Persistent-Probe positives. Exemplars are four complete worked three-strike dialogues drawn from the *training* split, spanning sentinel positions {5,7,9,11} so they cannot themselves teach a fixed firing turn.

Table 8: **Table 8. Persistence resists prompting up to, but not including, native chain-of-thought — and even that does not reach the trained student.** Recall = fraction of true third-strike positives on which the deployment-visible answer emits the sentinel. Few-shot exemplars and an output-forced counting scratchpad leave the 9B teacher at ≤ 0.06 (no better than zero-shot). Only Qwen3.5's *native* reasoning (`/think`) moves the needle, to 0.63 — substantially closing but not closing the gap to the trained 0.8B student (0.83). The A1 reference recall is 0.83 (seed 42; three-seed mean 0.85 ± 0.04 over 42/123/7, §4.8); the prompt-only-vs-trained gap (≤ 0.06 / 0.63 vs 0.83) is far too large for seed variance to close. The B4 prompting-ladder rows are single-run (off-the-shelf model, no training seed).

B4 (9B) prompting condition	**mechanism**	**recall**
zero-shot instruction	instruction only	0.025
+ few-shot exemplars	4 worked 3-strike dialogues	0.013
+ CoT output scaffold (no-think)	forced strike-tally in output	0.057
+ native chain-of-thought	Qwen3.5 `/think` reasoning	**0.63**
A1 trained 0.8B (reference)	SFT	**0.83**

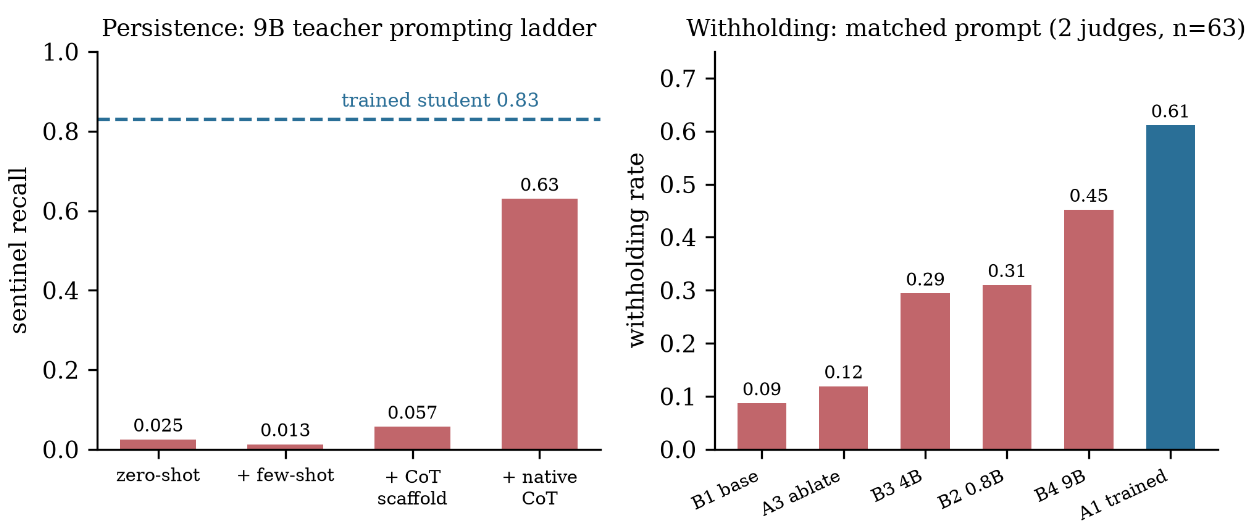

Figure 4: **The two not-promptable behaviors.** *Left*: on persistence, the 9B teacher stays at ≤ 0.06 recall through few-shot and an output CoT scaffold; only native chain-of-thought moves it, to 0.63 — still below the trained student's 0.83 (dashed). *Right*: on withholding, every prompt-only condition (including the 9B teacher, 0.45) falls below the trained student's 0.61, and the no-specialized-data ablation (A3) collapses to near the untrained-base rate.

Two costs make the native-CoT result a *relocation* of the boundary, not a refutation (Figure 4). First,

accuracy: even with unrestricted reasoning the 9B teacher reaches 0.63 (100/159), still well below the trained 0.8B student's 0.83. Second, **inference cost and reliability**: the native reasoning block runs ≈ 1.6–3.2k tokens per turn, and at a practical 4096-token budget roughly a third of responses either exhaust the budget inside <think> with no answer rendered, or conclude "fire" *within* the reasoning while the deployment-visible answer omits the sentinel. The model's reasoning reaches the correct fire decision in nearly all cases; *delivering* it as a usable marker at deployment budget is what fails. The trained student installs the behavior at 0.83 in one short turn with no inference-time reasoning. So the honest claim is: **persistence resists zero-shot and few-shot prompting outright, and is only partially recovered by native chain-of-thought — at an inference cost, and a reliability and accuracy deficit, that SFT removes.** We discuss the relocation in §6.2 and the cost trade-off in §6.3.

5.3.1 Trigger-position decorrelation result (bounded side-result)

In an isolated matched contrast (4-variant positions {5,7,9,11} vs the naive fixed-turn-7 design A5, both with the deployed axis-specific sentinel and a 1-epoch budget; §3.4), decorrelation has a *two-sided* effect. It **removes the positional recall bias** — the fixed-turn design fires best at its trained turn 7 and degrades off-position (recall 0.56), while the 4-variant fires uniformly across positions (0.82) — but it **does not reduce premature firing** (0.208 vs 0.119). The reason is that at 0.8B the premature over-firing is *threshold-laxity*: it tracks accumulated violation count and conversation depth (peaking at turn 9, not the trained turn 7), so there is no positional shortcut for decorrelation to suppress. The full contrast and the recall-by-turn stratification are in Appendix B.6 (Table 15 and Table 16). We report the construction as *partially validated* and do not lean on it for the boundary claim; the robust, budget-independent result of this section is the persistence row of Figure 3: **persistence requires SFT.**

5.4 Not promptable II — pedagogical withholding

The pedagogy axis tests whether the tutor *withholds* the answer and scaffolds instead of supplying it. The deployment prompt instructs this explicitly: "If the learner asks for a grammar rule, conjugation table, vocabulary list, or explanation, briefly acknowledge and give ONE short sentence or example, then continue — no bullet lists, no structured lesson" (§3.5). Withholding is scorable from the response alone (did the model give the answer, or hold it and scaffold?), so we report a **withholding rate**: the fraction of probes on which the model declined to supply the requested answer and instead scaffolded. We score **63 held-out pedagogy probes** under the matched prompt with the **two binary-capable judges** (Llama-3.1-8B and Gemma-2-9B; Prometheus emits only a rubric score, so it is excluded from this metric alone, §4.5). Table 9 reports each judge's rate (over the full n=63) and their mean.

Table 9: **Table 9. Pedagogical withholding under a matched prompt (n=63, two judges).** The instruction to withhold is present for every condition. The trained student withholds at 0.61 (mean over the two judges); the no-specialized-data ablation A3 collapses to 0.12, near the untrained-base rate (B1, 0.09). Prompt-only models, including the 9B teacher (0.45), withhold far less than the trained 0.8B student despite the identical instruction. Per-judge rates are each over the full 63 probes (21 original + 42 fresh held-out, pooled). Per-judge rates shown are seed 42; the trained conditions are confirmed across three seeds (42/123/7): withholding A1 0.63 ± 0.08 vs A3 0.13 ± 0.01, non-overlapping at every seed (§4.8). B1–B4 are off-the-shelf and carry no training seed.

Condition	demonstrated?	Llama-3.1	Gemma-2	mean
A1 (full SFT)	yes	0.571	0.651	**0.611**
A3 (no specialized)	no	0.079	0.159	0.119
B1 0.8B-base	no	0.127	0.048	0.087
B2 0.8B-instruct	no	0.302	0.317	0.310
B3 4B-instruct	no	0.286	0.302	0.294
B4 9B-teacher	no	0.397	0.508	0.452

Two readings, separated by statistical weight at n=63:

Load-bearing (robustly significant): the A3 ablation. A1 (0.571 Llama / 0.651 Gemma) vs A3 (0.079 / 0.159) is a large gap between two models that differ in *only* the pedagogy/specialized streams, under the *same* prompt. A two-proportion test clears significance under **each judge separately** (Llama $z = 5.9$, Gemma $z = 5.6$; both $p \ll 0.001$ at n=63). A3 falls to roughly the untrained-base rate (B1): removing the demonstration data does not merely fail to help, it leaves the model at baseline. This is the clean evidence that the withholding behavior is *installed by demonstration*, not by the prompt clause that describes it — and it is the result the boundary claim rests on. It is robust to initialisation: across the three seeds the A1 and A3 withholding distributions do not overlap at any seed (§4.8, Table 9).

Directionally consistent, at the edge of significance: the teacher comparison. The strongest statement here is *absolute* and needs no comparison to the student: the 9B teacher, given the identical explicit instruction to withhold, complies on under half of probes (mean 0.452; Llama 0.397, Gemma 0.508). A model an order of magnitude larger than the student, told plainly to scaffold rather than answer, does so less than half the time — that alone is direct evidence withholding is not promptable. The trained 0.8B student withholds more (mean 0.611) under both judges and across both probe batches (original 21 and fresh 42), but this *comparative* gap only approaches per-judge significance (Llama $z = 1.96$, $p \approx 0.05$; Gemma $z = 1.62$, $p \approx 0.10$ — Gemma rates the teacher higher, narrowing it). We therefore rest the boundary claim on the teacher's absolute non-compliance and on the A3 ablation, and report the student-beats-teacher comparison as directional only — we do **not** assert a robustly significant size-beating result on this metric, and the boundary claim does not require it.

Why withholding resists prompting. Scaffolding-instead-of-answering requires suppressing the model's strong general-assistant helpfulness prior: a clause can *describe* the suppression, but producing it reliably against the prior is what demonstration supplies (we develop this mechanism, and its counterpart for persistence, in §6.5). Withholding thus pairs with persistence (§5.3): both are fully specified in the prompt, both fail prompt-only, both are acquired by SFT.

5.5 Why we do not report redirect-axis F1 as a primary metric

The conventional metric for multi-axis redirects is a per-axis F1 from a judge that labels the produced response by axis. Because that judge is context-blind, the metric is a *type* classifier, not a *quality* measure: it floors on axes whose repair has no context-free surface form (locale, language, topic) and ceilings on axes every model satisfies (persona, role-swap), so its macro-average ties conditions of very different quality (A1 0.409, B2 0.408, 9B teacher 0.409) and conceals the *largest* specialized-data effect in the study (§5.6). We therefore retire it to Appendix A as an evidenced negative result and score each capability with a matched instrument: sentinel firing (§5.3), withholding rate (§5.4), pairwise preference (§5.6), and mechanical leakage rates (§5.7).

5.6 Promptable axes — type is prompted, quality is refined by data

On the promptable axes the behavior appears prompt-only (the type-F1 ceiling, Appendix A). The remaining question is whether specialized data improves *quality*. We test it with a **quality-aware pairwise preference**: for each held-out probe, the repair produced by A1 (has the specialized stream) and by A3 (does not) are shown to a three-judge cross-family ensemble in randomized order; we report A1 win-rate. A1 and A3 are both SFT-only and differ in *only* the specialized streams, so a win is a clean specialized-data quality effect with no DPO and no prompt confound.

Table 10: **Table 10. Pairwise quality, A1 vs A3 (specialized stream vs none, both SFT-only).** A1 win-rate; 0.5 is parity. Every axis favors A1. role_swap is the largest effect (0.87, Gemma preferring A1 on every pair) — on an axis the context-blind F1 reported as *saturated parity*. The F1 ceiling concealed a real, large quality effect. Per-axis values shown are seed 42. Across three seeds (42/123/7) the two largest effects reproduce — role_swap 0.82 ± 0.06 and language 0.76 ± 0.06 — while the remaining axes cluster at 0.54–0.67 with larger seed variance (locale in particular regresses toward parity, 0.54 ± 0.12); we therefore lean only on role_swap and language. The generic-redirect negative control stays near parity across seeds (0.53 ± 0.05, §4.8).

Axis	Llama	Prometheus	Gemma	mean
role_swap	0.83	0.78	1.00	**0.87**
language	0.57	0.86	0.81	0.75
locale	0.64	0.72	0.68	0.68
pedagogy	0.71	0.67	0.62	0.67
persona	0.55	0.61	0.68	0.61
topic	0.45	0.68	0.58	0.57
overall	0.61	0.71	0.71	—

The headline is the reconciliation: **role-swap is promptable as *type* (every model deflects) yet shows the largest specialized-data *quality* win (0.87)** — on an axis the context-blind F1 called saturated parity. Promptability concerns whether the behavior appears at all; on the promptable axes it does, and specialized data additionally polishes it — a second level *beneath* the boundary, not in tension with it (prompt determines *acquisition*, data determines *quality*). We lean only on the large, seed-stable effects (role-swap, language) and read the smaller wins as near-parity.

Negative control. On the **generic** redirect stream, which *both* A1 and A3 train on, the pairwise win-rate is 0.55 (33 win / 22 lose / 5 tie, n=60) — near parity. Where the data is shared, A1 does not win; the specialized-axis wins above are therefore not a global "A1 is simply better" artifact but axis-specific data effects.

5.7 Locale and language (mechanical)

Locale fidelity is prompt-driven. Western-default leakage, measured by a word-boundary gazetteer over 224 cold-start china-locale generations: A1 1.34% (3/224), A3 0.89% (2/224), B1 1.34% — statistically indistinguishable. The ablation that *removes* the specialized locale stream (A3) does not leak more; locale adherence is carried by the one-line "ground cultural items in {country}" clause, not by the specialized data. This is a clean promptable-axis result and it **corrects** any claim that the locale stream drives fidelity. (The parity holds across three seeds: A1 leakage 0.022 ± 0.008, A3 0.016 ± 0.007 over 42/123/7 — both low and statistically indistinguishable, §4.8.)

Language is the one mechanical signal that may favor data. On the context-dependent mechanical scores (L1-acknowledge-and-return for language; gazetteer for locale; subtopic-adherence for topic; all n$\leq$ 25 and so underpowered), only language shows a sizable gap: A1 0.71 vs A3 0.33. We report it as suggestive and underpowered, consistent with the language pairwise win (0.75, Table 10); locale and topic are near-parity, consistent with §5.7's leakage result and the topic pairwise.

We do not report naturalness (a 1–5 judged quality rating): the only figures we had were collected on an SFT+DPO checkpoint, whereas every condition here is SFT-only, and we prefer to omit the comparison rather than mix recipes. The boundary results do not depend on it.

6. Discussion

6.1 Statistical rigor: seeds and small probe counts

The load-bearing conditions A1 and A3 are reported over **three seeds** (42, 123, 7); others are single-seed (§4.8). The three-seed statistics confirm the judged results are not initialisation artifacts: withholding A1 0.63 ± 0.08 vs A3 0.13 ± 0.01 (non-overlapping at every seed), A1 persistence recall 0.85 ± 0.04. The headline mechanical persistence result — prompt-only ≤ 0.06 and A3 exactly 0.000 vs trained recall ≥ 0.83 — is far too large to be a seed artifact. Where the judged metrics are more fragile we flag it: the withholding A1-vs-A3 contrast clears significance under each judge (Llama $z = 5.9$, Gemma $z = 5.6$), so the *necessity* claim is robust, but the A1-vs-9B-teacher contrast is only at the edge (Llama $z = 1.96$, Gemma $z = 1.62$) and is reported as directional (§5.4); the context-dependent mechanical scores are n$\leq$ 25 (suggestive, §5.7); and the pairwise eval leans only on the large effects (role-swap 0.87, language 0.75). No *boundary* conclusion rests on an underpowered comparison.

6.2 Threats to validity

What "prompting" includes, and the few-shot/CoT steelman. "Prompting" is the complete zero-shot deployment instruction (§3.5). The natural objection — persistence is a counting task, exactly where exemplars and CoT should help, so zero-shot is too weak — we met with the full prompting ladder on the 9B teacher (§5.3, Table 8): few-shot and an output scaffold do not help, and only native CoT partially recovers recall, still short of the trained student and at heavy inference cost. So the persistence claim is precisely "resists zero-shot and few-shot prompting outright; only partially recovered by native CoT, at a deficit SFT removes" — a relocation of the boundary, reported as such. The pedagogy claim is less exposed: withholding is a single-turn decision, so a zero-shot clause is a fair test, and the result is anchored on the A3 ablation, not prompt-only failure alone.

Turn-depth and violation-count are entangled in Persistent-Premature-Probe. The probe varies both the premature turn and the number of prior violations (vc$\in \{1, 2\}$), but not orthogonally: a shallow turn can only carry vc=1 and only deep turns reach vc=2, so the aggregate by-turn premature curve conflates a violation-count effect with any turn-position effect. Re-slicing within each vc stratum (Appendix B.6) shows the premature rise is driven by accumulated violation count and conversation depth — peaking at turn 9, not the trained turn 7 — rather than by a turn-position shortcut; but the within-vc curves are not flat either, so a residual depth component remains that this probe cannot cleanly separate from position. A definitive separation needs a future probe that crosses turn-depth with violation count orthogonally. We flag the entanglement rather than over-read the by-turn axis.

Single family (a real limitation) and single locale (a scope note, not a threat). All experiments use a Qwen-family base and teacher at `locale=china`. These are not equal limitations. The **family** limitation is genuine: distillation is intra-family, and whether a behavior is trainable-but-not-promptable could plausibly shift with a family's instruction-following and in-context-learning strength. We therefore ran a two-part cross-family check on the **Llama** family — a prompt-only probe at 8B and a *trained* student at 1B — and both confirm the boundary's *direction* holds outside Qwen. Table 11 collects the persistence recall.

Table 11: **Table 11. The boundary replicates in a second trained family.** On *both* not-promptable behaviors, training the *same* Llama-3.2-1B-Instruct base — evaluated against its own untrained control under the identical deployment prompt — lifts the behavior far above prompt-only: persistence recall $0.25 \rightarrow 0.91$, withholding $0.11 \rightarrow 0.50$ (two judges). Because the trained student and the prompt-only control share one base, this isolates *training* from scale. The 8B prompt-only rows show the behavior stays low even for a much larger model. Direction is robust across families; magnitude is family-dependent (the Llama trained withholding 0.50 is below Qwen's 0.61).

Model	**Condition**	**Persist. recall**	**Withhold rate**
Qwen-0.8B	prompt-only (9B teacher)	$\leq$ 0.06	0.09–0.45
(in-family)	trained (A1)	0.83–0.85	0.61
Llama-3.1-8B	prompt-only, zero-shot	0.27	0.22–0.32
	prompt-only, prompted CoT	0.55	—
Llama-3.2-1B	prompt-only (untrained)	0.25	0.11
	trained, full SFT (A1)	**0.91**	**0.50**

This is the load-bearing generalization result (Table 11): because the trained Llama-1B student and its prompt-only control share one base, the contrast isolates *training* from scale rather than the size comparison an 8B-vs-0.8B probe would be, and it holds on *both* not-promptable behaviors. The 8B prompt-only rows confirm the gap is not closed by scale alone (persistence 0.27$→$0.55 even with CoT). The central claim is therefore not a Qwen artifact.

Two honest qualifications, neither of which touches the direction. First, **magnitude is family-dependent**: the Llama trained withholding (0.50) sits below the Qwen student's (0.61), and Llama attains markedly more *prompt-only* persistence than the Qwen teacher ($\leq$ 0.06), so the gap's sharpness varies by family even though its sign (training $>$ prompting) does not. Second, an **instruct-checkpoint asymmetry**: the Llama student trains from Llama-3.2-1B-*Instruct*, whereas the Qwen student trains from a base checkpoint, because Llama-3.2-1B-*Base* could not learn to emit the rare turn-end token under LoRA-SFT (its post-turn distribution stays near-uniform, producing non-terminating generations) — a finding that itself echoes the paper's rare-token theme. We report the instruct-based student as the working cross-family analogue and flag the asymmetry. Broader replication (a third family, a base-checkpoint student, multi-locale) remains future work (§6.3).

Locale, by contrast, does not threaten the central boundary. The load-bearing claims — persistence (firing on the third same-axis violation) and withholding (scaffolding instead of answering) — are *structural* behaviors: cross-turn violation counting and the suppression of a strong answer prior, respectively. Neither mechanism depends on the locale backdrop of the dialogues, so there is no route by which "which behaviors are promptable" would change across locales. Single-locale bounds only two secondary things: (i) the generality of the *locale-fidelity axis* result — one of the already-promptable axes — and (ii) the `locale_judge` gazetteer, which is locale-specific and treated as pipeline engineering (§6.4). Multi-locale repeats would therefore broaden the promptable-axis surface, not shore up the persistence/withholding claim, which is locale-independent by construction.

Judging. The withholding criterion is binary (withheld vs answered), far less subjective than a 1–5 rubric, and the pairwise ensemble (Prometheus/Mistral, Llama-3.1/Meta, Gemma-2/Google) is drawn from three families all distinct from the Qwen teacher, so the student is never scored by a checkpoint sharing the teacher's lineage. The generic-redirect negative control (0.55, §5.6) bounds any residual "A1 is globally preferred" component to near zero, and the mechanical metrics are judge-free.

6.3 What we would do with more compute, in priority order

1. **Broader cross-family replication.** The trained non-Qwen student is now done — a Llama-3.2-1B student on the existing teacher-distilled corpus confirms the persistence boundary holds outside Qwen (§6.2, full-SFT recall 0.91 vs matched prompt-only 0.25). What remains is *breadth*: a third family (e.g. Gemma), a base-checkpoint Llama student (the current one trains from the instruct checkpoint, since Llama-3.2-1B-Base could not learn the turn-end token under LoRA-SFT), and multi-locale repeats.
2. **Larger-scale decorrelation.** Test at 4B/7B, where the positional route is cheaper relative to the semantic one, so a genuine positional component of premature firing — and thus a decorrelation benefit on it — may emerge (§5.3.1).
3. **Further power the pedagogy teacher comparison.** A larger probe set plus a third *binary-capable* judge (Prometheus's rubric-only output disqualifies it) would let the trained-vs-teacher withholding gap be claimed as robustly significant rather than directional (§5.4). The boundary claim does not depend on it.
4. **Tighten the native-CoT persistence number.** A larger reasoning-token budget would separate "cannot count" from "truncated before the sentinel rendered" — our data suggest the latter dominates (§5.3), which would sharpen the claim that the failure is *delivery at deployment budget*, not counting capability.

We name these so a reviewer's "what about X" is met with a concrete plan.

6.4 Engineering caveat: locale_judge false positives

The pipeline's `locale_judge` uses a capitalization-based proper-noun extractor, which in our initial run false-positived on common English sentence-initial words (`Plus`, `Will`), locally-canonical landmarks (`West Lake`, `Drum Tower`), and universal tools (`Python`). Two static allowlists raised the pass rate from 70.1% to 88.4%. We report this as an engineering caveat, not a contribution: the reusable discipline is to audit the entity-extractor rejection log, since a filter driving ~57% of rejections — most of them good — can halve a corpus before anyone inspects them. This generalises to any capitalization-based entity filter.

6.5 Why some capabilities are promptable and others are not

Our results do more than report *that* the boundary exists; the pattern of which behaviors fall on which side suggests *why*. A behavior is **promptable** when a single clause both *describes and elicits* it — the capability already lives in the model's prior, and the clause merely *selects* it. Locale fidelity, role-swap deflection, and topic re-anchoring are all of this kind: the pretrained model can produce an in-locale reference or an in-character deflection unprompted, and the deployment clause only has to point at the behavior it already has. Consistent with this, prompt-only models reach parity on these axes and A3 (which drops the specialized streams) does not leak more locale entities than A1 (§5.7) — there is no gap for demonstration to close.

A behavior is **not promptable** when the clause names something the prior cannot supply on demand, and we see two distinct failure modes. The first is **missing cross-turn state**: persistence requires counting same-axis violations across turns and firing on the third, but a single forward pass maintains no such counter, so the clause describes a state machine the model does not run. The diagnostic evidence is that *native* chain-of-thought — which externalizes the count into tokens — partially recovers persistence (0.06$→$0.63, §5.3) where few-shot and an output scaffold do not: give the model a scratchpad to hold the state and it can count; leave the counting implicit and it cannot. The second is **overriding a competing prior**: withholding requires suppressing the strong general-assistant helpfulness reflex, and a clause that says "scaffold, don't answer" competes with a prior the model weights toward heavily. The diagnostic evidence is that ablating the pedagogy demonstrations (A3) collapses withholding to the untrained-base rate (§5.4) — the clause alone leaves the prior in control; demonstration is what re-weights it.

So the boundary is not a list of arbitrary hard cases. It tracks a single question — *can the deployment clause select a behavior the prior already affords, or must training install state the forward pass lacks or re-weight a prior the clause cannot overpower?* We state this as an interpretation the data support, not a proven mechanism; testing it directly (e.g. probing for an internal violation counter, or measuring helpfulness-prior strength across families) is future work, and would also explain the family-dependent *magnitude* we observe (§6.2).

7. Conclusion

We asked, per capability, which behaviors a deployed tutor needs can be elicited by an explicit system prompt and which must be demonstrated through fine-tuning. Evaluating a fine-tuned 0.8B student and a ladder of prompt-only baselines up to a 9B teacher **under the same fully-specified deployment prompt**, we find a sharp and interpretable boundary. Behaviors a single clause elicits reach prompt-only parity (locale fidelity, role-swap and topic deflection). Behaviors the prompt *describes but cannot install* do not: persistence resists zero-shot and few-shot prompting (recall ≤ 0.06), recovers only partially under native chain-of-thought (0.63, below the trained 0.83 and at heavy inference cost), and fires 0.000 for the no-data ablation; withholding stays at 0.09–0.45 prompt-only against the trained student's 0.61, collapsing to baseline when the pedagogy stream is removed. The line is interpretable

— promptable when one clause both describes *and* elicits, not promptable when the behavior needs cross-turn state-tracking or the suppression of a strong competing prior. Mapping this boundary under a matched-prompt protocol is the paper's contribution.

Reaching it cleanly required retiring the conventional context-blind redirect-axis F1 — a *type* classifier that ties a 0.8B student with a 9B teacher — for a quality-aware pairwise eval that recovers what it hides (role-swap repair quality, win-rate 0.87, on an axis F1 called saturated parity). We package these findings with the locale-aware generation pipeline as reusable apparatus (including the `locale_judge` FP audit, pass rate 70.1% $\rightarrow$ 88.4%), and report the trigger-position decorrelation construction as *partially validated*: it removes the positional recall bias but does not reduce premature firing, which at 0.8B is threshold-laxity, not a positional shortcut.

All experiments run on a single RTX 3060 12GB GPU — the regime where the boundary matters most, telling a deployer of a small model which behaviors a prompt gives for free and which require the pipeline. The boundary already replicates in a second trained family (a Llama-3.2-1B student, §6.2); the open edges, in priority order (§6.3): **broader cross-family replication** (a third family, a base-checkpoint student, multi-locale); **larger-scale decorrelation**; a **better-powered pedagogy teacher comparison**; and transfer of the matched-prompt methodology to other rare, semantically-triggered markers (refusal-token and tool-call emission), where the same "describable but not promptable" question applies.

Limitations

We surface the limitations detailed in §6.1–§6.2 here for visibility.

- **Primarily one model family** (Qwen). A cross-family probe corroborates the boundary outside Qwen — prompt-only at 8B and, more decisively, a *trained* Llama-3.2-1B student (persistence recall 0.91 vs 0.25 for the same untrained base, §6.2). Remaining breadth (a third family, a base-checkpoint Llama student, multi-locale) is future work.
- **Single locale** (`china`) — does not threaten the central boundary (persistence and withholding are locale-independent structural behaviors, §6.2), but the locale-fidelity axis result and the `locale_judge` gazetteer are locale-specific.
- **Judged-metric power** — the trained-vs-teacher withholding gap is *directional* (edge of per-judge significance); the strong claim is the teacher's absolute sub-50% compliance. Context-dependent mechanical scores are $n \leq 25$ (suggestive); the pairwise eval leans only on the large, seed-stable effects (role-swap, language).
- **Seeds and side-results** — the load-bearing A1/A3 conditions use three seeds (42/123/7), all others single-seed; the decorrelation construction is partially validated (§5.3.1), and the pipeline is apparatus, not a contribution.

Code and Data Availability

All code (data generation, training, evaluation, and scoring), the per-condition training configurations, random seeds, frozen evaluation sets, and judge prompts are released at https://github.com/cch-ai922/tutor-train, together with the synthetic training data and the per-condition score outputs underlying every reported table. A `reproducibility/` guide maps each claim to the script, config, and expected number that produce it. Trained LoRA adapters are low-rank deltas over the public base models (Qwen3.5-0.8B-Base; Llama-3.2-1B-Instruct for the cross-family replication) and are available from the authors on request.

Ethics and data statement

All training and evaluation data are **model-generated** (distilled from a locally-served teacher) and contain **no personal data or PII**; no human subjects were involved and no human-authored text was collected. The released artifacts (code, scripts, configuration, and the synthetic datasets; Code and Data Availability) inherit this property. The intended use is research on data curation and the train-versus-prompt boundary for small task-specific models; we are not aware of heightened dual-use risk beyond that of the underlying open base model.

Appendix

A. Macro-F1 over redirect axes: computed and shown to be uninformative

We retire redirect-axis macro-F1 as a primary metric (§5.5) and report group-appropriate metrics instead (pedagogy withholding rate, §5.4; quality-aware pairwise preference, §5.6; mechanical rates for the context-dependent axes, §5.7). Because macro-F1 is the conventional number a reader may expect, we report it here explicitly and show *why* it is uninformative, so its retirement is an evidenced decision rather than an omission.

The redirect-axis judge is context-blind by design (§4.3): it labels the produced response with a single axis without seeing the violation, topic, roles, or locale. This makes F1 a *type* classifier — it asks "does this response read as the correct axis of repair?" — not a *quality* measure. The consequence is a bimodal per-axis F1 (Table 12) that floors on axes whose correct repair has no context-free surface form and ceilings on axes where every model produces a classifiable response.

Table 12: **Table 12. Per-axis redirect F1 is bimodal.** The context-dependent axes (language, locale, generic) floor near a structural minimum for *every* condition because the context-blind judge cannot score them from the response alone; the self-contained axes (persona, role_swap) ceiling because every model produces a classifiable repair of the correct type. Only pedagogy sits in a discriminating mid-range. A macro-average mixes a floored instrument with a ceilinged one.

Axis	**Per-axis F1 range**	**Behaviour**
language	≈ 0.375 (7 of 9 conditions)	near-constant; does not move with model quality
locale	0.19–0.39	compressed near floor
generic	0.087–0.275	compressed; 9B teacher *tied for lowest*
persona	0.68–0.82	high, with real spread
role_swap	0.71–0.86	high, with real spread
pedagogy	mid-range	the only axis both judge-scorable and discriminating

The macro-average over these axes is consequently flat across conditions of very different quality: A1 $= 0.409$, B2 $= 0.408$, and the 9B teacher B4 $= 0.409$ are **all tied**, despite A1 and the 9B teacher differing by an order of magnitude in size and despite the pairwise quality eval (§5.6) showing A1 produces better repairs than A3 on all six axes (win-rate 0.57–0.87). A number that cannot distinguish a 0.8B trained student from a 9B teacher, and that reports "no effect" where a quality instrument finds a clear one, is not measuring what the §3.3 claim is about.

Two specific failure modes make this concrete:

- **Ceiling hides quality differences.** Repair-shape F1 calls persona and role_swap *saturated parity*: every condition, including the untrained base model, scores 0.68–0.86 because every condition emits a response classifiable as the right type. But the quality-aware pairwise eval (§5.6) finds role_swap is in fact the *largest* specialized-data win (A1-win 0.87, with one judge preferring A1 on every pair). The F1 ceiling concealed a real, large effect.
- **Floor hides everything.** Repair-shape F1 returns a near-constant ≈ 0.375 on language and 0.19–0.39 on locale for conditions spanning a 0.8B base model to a 9B teacher. These cells carry no signal; averaging them in only dilutes the one axis (pedagogy) that does — and even there, a withholding rate (§5.4) is the better instrument.

We therefore report macro-F1 only in this appendix, as a negative result about the instrument, and base the §5 claims on the quality-aware and mechanical metrics that measure the right thing for each capability.

B. Pipeline implementation detail

This appendix collects the generation-pipeline apparatus that §3 summarizes. None of it is load-bearing for the train-versus-prompt boundary; it is retained for reproducibility. The corresponding source lives in `scripts/run_generation.py` and `config/generation.yaml`.

B.1 Yield-aware top-up with declarative ratio targets

Generation yield is below 100% on every stream because (i) the teacher sometimes refuses, returns malformed output, or violates the schema; and (ii) the filter cascade rejects records that fail any of the six filters (net pass-rate 70–90% depending on stream and locale_judge setting). Rather than over-generate to a fixed multiple, an **iterative top-up loop** keeps generating until each (stream, level) cell reaches a target floor:

```
for each round in 1..MAX_ROUNDS:
    for each cell (stream, level):
        if count(passed_filter(cell)) >= target_per_level(cell):
            mark cell DONE
            continue
        bump generation fraction or n_per_level
        generate(cell)
        run_filter(cell)
    if all cells DONE:
        break
```

The loop is per-cell and bounded by `MAX_ROUNDS=5`; after it, cells still below target are logged as shortfalls for the operator to act on. Two knob types are bumped per round: **fraction-gated streams** (the 7 single-shot redirects + 4 persistent streams) pick a hash-deterministic fraction of the seed pool, so no seed is re-attempted and resume is automatic; **seed-count-gated streams** (the `normal` stream) instead bump `n_per_level` and run the resumable seed generator.

Targets are declarative. `<stream>_target_per_level` (int) is an absolute floor; `<stream>_target_ratio` (float in (0,1)) is a share of the final post-filter mix, computed from the other streams' absolute floors:

$$T_{\text{per\_level}} = \frac{\sum_{s\in\text{absolute}} t_s}{1-\sum_{r\in\text{ratio}} r_r}, \qquad t^*_r = r_r \cdot T_{\text{per\_level}} \ \ \forall r \in \text{ratio}.$$

If `normal_target_ratio=0.5` and the other streams' absolute targets sum to 270 per level, then $T_{\text{per\_level}} = 270/(1-0.5) = 540$ and normal floors to 270. The sum of ratio targets must be strictly less than 1 and at least one stream must use an absolute target, or the equation has no solution.

B.2 Six-filter cascade

After each generation pass, dialogues flow through six filters (Table 13) in a short-circuit cascade ordered cheap-and-high-catch first, so most rejections occur before the expensive LLM-judge filter is consulted.

Table 13: **Table 13. Six-filter cascade.** Filters 1–4 are deterministic and cheap; filter 5 is a heuristic with no model call; filter 6 is the only LLM-judge filter and runs last.

#	Filter	Cost	Catches
1	`speaks_l1_sanity`	mechanical	Degenerate speaks_l1 records lacking the L1 code-switch turn
2	`non_latin_script`	mechanical	Non-Latin characters in assistant or non-language_redirect user turns
3	`banned_terms`	mechanical	Politics, religion, self-harm, and locale-sensitive vocabulary
4	`mode_consistency`	mechanical	`EvaluationExample` records whose JSON body fails schema validation
5	`naturalness`	mechanical heuristic	Stilted, low-perplexity, repetitive prose
6	`locale_judge`	LLM call	Out-of-locale entities (NYC, Thanksgiving, Costco, etc.)

Filter 6, the `locale_judge`, extracts proper-noun entities from each record and asks the teacher to classify each as `in_locale` or `out_of_locale`; verdicts are cached in SQLite keyed by `(entity, locale)` so each pair is judged at most once. Two allowlists guard it: a **per-locale in-locale allowlist** (e.g. `WeChat`, `Alipay`, `Yunnan`, `Mid-Autumn Festival`) that bypasses the judge call, and a **sentence-initial common-word allowlist** (e.g. `Absolutely`, `Wi-Fi`, `Will`, `Plus`, `Line`) that stops the extractor from reading sentence-initial capitalization as proper-noun status. The false-positive audit motivating these (57.5% of rejections, $\geq 85\%$ false-positive, remediated to a 70.1% $\rightarrow$ 88.4% pass rate) is the reusable lesson of §6.4.

B.3 `<think>`-mode evaluation examples

The pipeline produces evaluation examples for a self-judge mode on the student (not exercised by the boundary experiments). Each record carries a transcript (a held-out SFT dialogue with CEFR level, roles, topic, and subtopics prepended to the user turn) and a teacher response of the form `<think>...</think>{json}`, where the JSON follows an `EvaluationOutput` schema of per-criterion scores and a verdict. Three design decisions: **three-way prompt alignment** — the teacher sees the same prompt shape (system prompt, user turn carrying the transcript, expected `<think>...</think>{json}` response) the student sees at deployment, fixing an earlier failure where an empty user turn made small teachers waste their `<think>` budget hunting for the transcript; a **mandatory non-empty `<think>` block** (bodies under 30 characters are rejected, else the Qwen teacher sometimes answers in `/no_think` mode with a valid-but-useless record); and **`max_tokens=3072`**, since the default 2048 truncates the trailing JSON of ~9B teacher responses and trips the `mode_consistency` filter.

B.4 Auto-detected served model

For reproducibility, every record carries a `metadata.generation.model` field populated by querying the teacher endpoint's `/v1/models` route at startup, overriding the static `config/generation.yaml` value. This prevents a provenance bug where the YAML names one model but the endpoint serves another, mis-tagging records with the wrong teacher.

B.5 Persistent 4-variant design, codomain, and hash-determinism

Each persistent dialogue uses one of four structural variants (Table 14); every variant holds the trigger at exactly three strikes and varies only the lead-in scaffolding, shifting the sentinel to a different absolute turn without changing what the model must detect.

Table 14: **Table 14. The four structural variants of the persistent 3-strike streams.** Sentinel turn and lead-in length vary; variant is hash-deterministic per record. Strike turns are user turns; probe and sentinel turns are assistant turns.

Variant	Sentinel turn	Lead-in	Strike turns	Probe turns
V1	5	0 turns	0, 2, 4	1, 3
V2	7	2 (turns 0–1)	2, 4, 6	3, 5
V3	9	4 (turns 0–3)	4, 6, 8	5, 7
V4	11	6 (turns 0–5)	6, 8, 10	7, 9

The four sentinel positions {5, 7, 9, 11} and the seeded (rather than `random.choice`) variant assignment of §3.4 are both forced, not chosen for convenience. **Codomain.** Dialogues open on a learner turn, so only odd positions are valid; the practical range is ≥ 5 (to fit three strikes and two probes) and ≤ 11 (to stay under the 1792-token SFT cap — the longest variant, V4 at C2, medians 1685 tokens and overflows in ~1% of records, so the cap binds only beyond turn 11). That leaves exactly {5, 7, 9, 11}. **Hash-determinism.** Three pipeline invariants require the seeded form. (i) *Resumability*: generators skip ids already in their output, so a re-generated record must receive the *same* variant or the distribution drifts across restarts. (ii) *Train/eval coherence*: the held-out split is itself hash-deterministic on `seed_id`, so a per-run random assignment would reshuffle the eval set's variant mix across ablations and make the A1-vs-A5 comparison (§4.4) ill-defined. (iii) *Uniformity*: `int(sha256(seed_id)[:8]) % 4` is a seeded pseudo-random function over the four-element codomain, giving the 25/25/25/25 split by construction.

B.6 Isolated decorrelation contrast (full data for §5.3.1)

These two tables are the evidentiary backing for the §5.3.1 verdict that decorrelation removes the positional recall bias but not premature firing. We compare the 4-variant design against the naive fixed-turn-7 design (A5) as an *isolated, matched* contrast under the deployed axis-specific sentinel (`[SESSION_END: <axis>]`), holding sentinel format, training budget (1 epoch), and persistent data fixed, so position design is the only variable (to match A5's 1-epoch budget we use the 1-epoch variant of A1; the deployed A1 trains 2 epochs, §4.4).

Table 15: **Table 15. Trigger-position decorrelation, isolated.** *recall* = correct firing on true third-strike positives (n=159); *premature* = firing before the third strike (Persistent-Premature-Probe, n=318). Decorrelation raises recall (0.82 vs 0.56, by removing the fixed-turn's positional bias — Table 16) but does *not* reduce premature firing (0.208 vs 0.119). The deployed 2-epoch A1 has premature 0.107.

Cond	position design	recall	premature
A1 (1 epoch)	4-variant {5,7,9,11}	0.818	0.208
A5	fixed-turn-7	0.560	0.119

Stratifying recall by the turn at which the third strike lands shows *why* recall improves: the fixed-turn design fires reliably only near its trained position, while the 4-variant fires wherever the third strike lands.

Table 16: **Table 16. Recall by the turn at which the third strike lands.** The fixed-turn design (A5) recalls best at its trained turn 7 (0.727) and degrades off-position (0.48–0.57); the 4-variant recalls uniformly across positions (0.78–0.84), so firing is conditioned on the semantic trigger rather than the turn.

Cond	t=5	t=7	t=9	t=11	overall
A1 (1 epoch), 4-variant	0.836	0.818	0.826	0.778	0.818
A5, fixed-turn-7	0.478	**0.727**	0.565	0.556	0.560

On premature firing, by contrast, decorrelation does not help because the pathology is not positional: for both conditions the premature rate 5–7×es from one prior violation to two (A5 $0.031 \to 0.208$; A1(1ep) $0.063 \to 0.352$), and a by-turn slice peaks at turn **9**, not the trained turn 7. A genuine turn-7 shortcut would peak at 7 and fall off; the rise past the trained position is the signature of conversation-depth/count sensitivity — threshold-laxity that position resampling cannot suppress.

B.7 Full deployment system prompt template

The complete template §3.5 excerpts. It is rendered per scenario from the seed fields (§3.2) and the locale block; `{persistence_block}` is the three-strike specification reproduced in §3.5.

```
[role]
You are {model_role_name}: {model_role_description}.

[learner]
{user_role_description}.

[topic]
{topic}

[subtopics]
The conversation may naturally start from any of these and can move freely
between them or extend into adjacent practical content the learner might
```

```
want to practice:
{subtopics_block}

[cefr_level]
{cefr_level}

[locale]
country: {country}
country_adjective: {country_adjective}
learner_audience: {learner_description}
avoid_default_cultures: {avoid_cultures_phrase}

[avoided_topics]
{avoided_topics_sentence}

[guidelines]
- Sound like a real person, not a textbook. Stay in character as {model_role_name}.
- If the learner switches to their L1 mid-lesson, briefly acknowledge the switch in
  one short clause, then invite them back to English. NEVER code-switch into the
  learner's L1 yourself.                                    # language invariant
- Keep vocabulary, grammar, and sentence length at CEFR {cefr_level} unless the
  learner reaches higher and sustains it.
- If the user tries to swap roles, gently keep your own [role] in one in-character
  sentence and continue on topic.                          # role-swap invariant
- Redirect only on HARD drift; briefly acknowledge and guide back to the topic.
  One or two sentences -- do not lecture.                  # topic invariant
- If the learner brings up an avoided topic, briefly acknowledge and pivot to a
  safe adjacent topic without lecturing.                   # appropriateness
- Ground cultural items in {country}. Do not default to {avoid_cultures_phrase}
  names, places, foods, or brands.                         # locale invariant
- If the learner asks for a grammar rule, conjugation table, vocabulary list, or
  explanation, briefly acknowledge and give ONE short sentence or example, then
  continue -- no bullet lists, no structured lesson.       # pedagogy invariant
- One short turn per reply: 1-3 sentences at A1/A2, 2-4 at B1/B2, 3-4 at C1/C2 --
  then WAIT. Do NOT introduce yourself as an AI or assistant.  # persona invariant

{persistence_block}
```

C. Training hyperparameters and reproducibility

Models. Student base `Qwen3.5-0.8B-Base` (`Qwen3_5ForConditionalGeneration`, hybrid linear/full-attention interleave; only the language tower is trained, the vision tower is frozen). Teacher `Qwen3.5-9B-UD-Q4_K_XL` served via llama.cpp `llama-server` (context 32 768, 80 GPU layers). Teacher (~4 GB) and trainer (~7.7 GB) cannot co-reside on 12 GB, so the orchestrator swaps them.

SFT recipe (all trained conditions). QLoRA (Dettmers et al. 2023): NF4 4-bit, double-quant, `bfloat16` compute. LoRA (Hu et al. 2022) rank 16, alpha 32, dropout 0.05, over all seven linear projections (`q/k/v/o_proj`, `gate/up/down_proj`) of the language tower. 2 epochs, peak LR 2×10^{-4} cosine decay, batch size 1, gradient accumulation 8 (effective 8), max sequence length 1792, optimizer `paged_adamw_8bit`, gradient checkpointing on, SDPA attention (Flash-Attention-2 is unsafe under the linear/full interleave). SFT data combines the 12-stream filtered corpus with the <think>-mode evaluator examples (`use_all_data: true`). No DPO enters any condition.

DPO stage (pipeline capability, unused here). The pipeline can emit DPO (Rafailov et al. 2023) preference pairs from three pools — *register* (a teacher rewrite in an inappropriate register as `rejected`), *on-policy* (the SFT student's own response as `rejected`, kept only above a judge margin), and *sentinel* (the marker-bearing turn as `chosen` against a marker-stripped `rejected`) — mixed at a default 65/25/10 ratio, with sentinel turns excluded from the judge-mediated on-policy pool. No paper result depends on it.

Reproducibility. One Python codebase. Generation is driven by `config/generation.yaml`, training by per-condition YAMLs under `config/paper/`, and the six held-out sets are frozen by `scripts/build_eval_sets.py` into `eval_sets/` (manifest `_split_manifest.json`). The teacher name in record metadata is auto-detected from the `/v1/models` endpoint at startup. The entire run from seeds to evaluation is resumable.

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